project.org

PARENT RESOURCE GUIDE

Optical Illusions & Visual Deception

Resources to explore with your child at home

WELCOME

Resources to explore with your child at home. These books, videos, websites, and activities extend the learning from our Mini Museum exhibition and help build critical thinking skills about what we see and believe.

WHY THIS MATTERS

In an age of manipulated images, deepfakes, and AI-generated content, understanding how our eyes and brains can be fooled is more important than ever. These resources help your child move from "that's cool!" to genuine critical thinking about visual information—a skill they'll use for life.

GRADES K–5

For elementary students

BOOK

Duck! Rabbit!

By Amy Krouse Rosenthal. Is it a duck or a rabbit? This playful book teaches kids that the same image can be seen different ways—a perfect follow-up to the exhibition.

BOOK

Look! Look! Look!

By Tana Hoban. Die-cut pages reveal only parts of photographs, encouraging kids to guess before the full image is shown.

WEBSITE

National Geographic Kids: Brain Games

Interactive optical illusions and brain teasers. Explore together and see how easily both of your brains can be fooled!

BOOK

Can You See What I See? series

By Walter Wick. Hidden object photography books that train visual attention. Great for developing careful observation skills.

VIDEO SERIES

SciShow Kids: "Why Do We See Optical Illusions?"

Short, kid-friendly videos about how the brain works, including episodes on optical illusions and perception. Available free on YouTube.

ACTIVITY

Create Your Own Thaumatrope

Make the Victorian spinning toy that creates the illusion of two images merging. All you need is cardboard, string, and markers. Search "make a thaumatrope" for instructions.

GRADES 6–8

For middle school students

BOOK

The Invisible Gorilla: How Our Intuitions Deceive Us

By Christopher Chabris & Daniel Simons. Famous for the "invisible gorilla" experiment showing how we miss things right in front of us. Fascinating for family discussion about attention and perception.

DOCUMENTARY

"The Brain with David Eagleman" (PBS)

Episode: "What is Reality?" Explores how your brain constructs what you experience as reality. Watch together and discuss. Available on PBS streaming.

WEBSITE

Michael Bach's Optical Illusion Gallery

Over 100 interactive optical illusions with scientific explanations. Explore together at michaelbach.de/ot/

FAMILY ACTIVITY

Fact-Check Challenge

Pick a viral image or video from social media together. Use reverse image search, check Snopes.com or FactCheck.org, and see if you can determine if it's real or manipulated.

BOOK

A Beginner's Guide to Constructing the Universe

By Michael Schneider. The chapter on vision and geometry explains how eyes work and why we see patterns everywhere—great for math-curious kids.

TED TALK

"How We See the World" by Beau Lotto

A neuroscientist explains why your brain sees what it expects to see, with amazing visual demonstrations. Search on YouTube or TED.com—great for family viewing.

WEBSITE

Common Sense Media: Digital Literacy Resources

Lessons on identifying fake news, manipulated images, and deepfakes. Excellent for teaching your child to verify what they see online. commonsensemedia.org

GRADES 9–12

For high school students

BOOK

Thinking, Fast and Slow

By Daniel Kahneman. Nobel Prize winner explains how the brain makes decisions using two systems. Essential reading for understanding cognitive biases—great for college prep.

BOOK

The Invisible Gorilla

By Christopher Chabris & Daniel Simons. The psychology behind why we miss obvious things and how our intuitions about perception are often wrong.

TED TALK

"How False News Can Spread" by Noah Tavlin

TED-Ed animation explaining how misinformation spreads and why our brains are wired to believe convincing lies. Includes practical verification strategies.

BOOK

Deepfakes and the Infocalypse

By Nina Schick. A comprehensive look at AI-generated synthetic media and its implications. Important for teens navigating a world where seeing is no longer believing.

DOCUMENTARY

"The Social Dilemma" (Netflix)

Includes important sections on how algorithms manipulate what we see and how filtered reality affects decision-making. Consider watching together for discussion.

WEBSITE

Bellingcat's Online Investigation Toolkit

Tools and techniques used by professional investigators to verify images and videos. Learn how journalists fact-check in the digital age.

DISCUSSION PROJECT

Historical Visual Propaganda Analysis

Explore together how governments have used visual manipulation throughout history—posters, doctored photos, staged scenes. Compare to modern techniques.

MUSEUMS & ONLINE EXHIBITIONS

Museums, experiments & conversation starters

MUSEUM

The Exploratorium (San Francisco)

Interactive exhibits on perception and vision. Many activities available online at exploratorium.edu

MUSEUM

Museum of the Moving Image (NYC)

History of visual media from magic lanterns to modern cinema. Online exhibitions available.

MUSEUM

The Science Museum (London)

Extensive collection on the history of photography and optical devices. Online galleries available.

DIGITAL ARCHIVE

Library of Congress: Stereographs

Thousands of original stereoscopic images from the 1850s–1930s. loc.gov

HANDS-ON EXPERIMENTS TO TRY TOGETHER

01

The Color Afterimage Experiment

Stare at a bright red image for 30 seconds, then look at a white wall. You'll see a cyan afterimage—demonstrating how your eyes' color receptors get tired.

02

The McGurk Effect

Search YouTube for "McGurk Effect demonstration." Watch how what you SEE can change what you HEAR. Great conversation starter about how senses work together.

03

Create Your Own Zoetrope

Build a Victorian animation device using a coffee can, paper strips with sequential drawings, and a lazy Susan. Fun weekend project!

04

Reverse Image Search Practice

Take any viral photo and use Google Reverse Image Search to verify it. This is how journalists check images—teach your child this essential skill.

CONVERSATION STARTERS FOR HOME

- *"If your brain fills in your blind spots automatically, what other gaps might it be filling?"*
- *"When is it most important to verify what you see before believing it?"*
- *"What's the difference between a fun optical illusion and harmful visual manipulation?"*
- *"If seeing can no longer be trusted as evidence, what CAN we trust?"*